MASTER OF COMPUTER APPLICATIONS

PRACTICAL RECORD WORK
ON
20MCA243 – MOBILE APPLICATION DEVELOPMENT LAB

Submitted by

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DEPARTMENT OF COMPUTER APPLICATIONS

COLLEGE OF ENGINEERING VADAKARA

(CAPE – GOVT. OF KERALA)

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DEPARTMENT OF COMPUTER APPLICATIONS

COLLEGE OF ENGINEERING VADAKARA

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CERTIFICATE

Certified that this is the bonafide record work on the practical course **20MCA243 MOBILE APPLICATION DEVELOPMENT LAB** done and prepare **Mrs. AFSATH K H(Reg No.VDA25MCA-2003)**, 3rd Semester **MCA(2025-27 Batch)** student of Departmentof Computer Applications at College of Engineering Vadakara, in the partial fulfillment of the award of MCA Degree of APJ Abdul Kalam Technological University (KTU).

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EXPERIMENT NO 1

AIM:

Design a Login Form with username and password using LinearLayout and toast valid credentials.

SOURCE CODE

Activity Main.XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Username"
        android:textSize="21sp"
        android:layout_marginBottom="4dp" />
    <EditText
        android:id="@+id/user"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="text"
        android:layout_marginBottom="20dp" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password"
        android:textSize="21sp"
        android:layout_marginBottom="4dp" />
    <EditText
        android:id="@+id/pass"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:ems="10"
        android:inputType="textPassword"
        android:layout_marginBottom="8dp" />
```

```
 <Button
      android:id="@+id/btn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Login" />
</LinearLayout>
```

MainActivit.java

```
package com.example.login;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity
   { EditText user,password;
   Button btn;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);

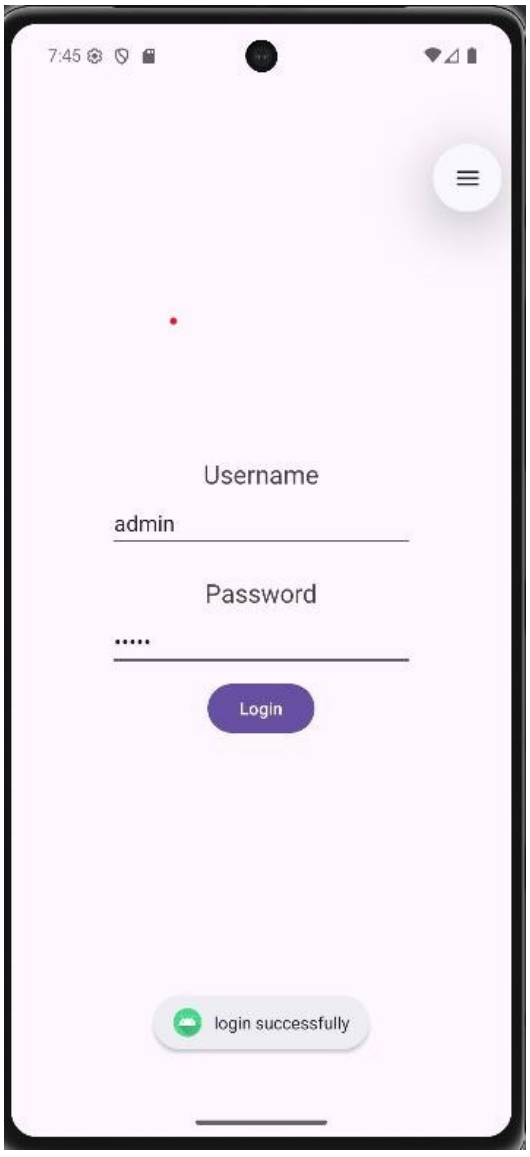
      user=findViewById(R.id.user);
      password=findViewById(R.id.pass);
      btn=findViewById(R.id.btn);

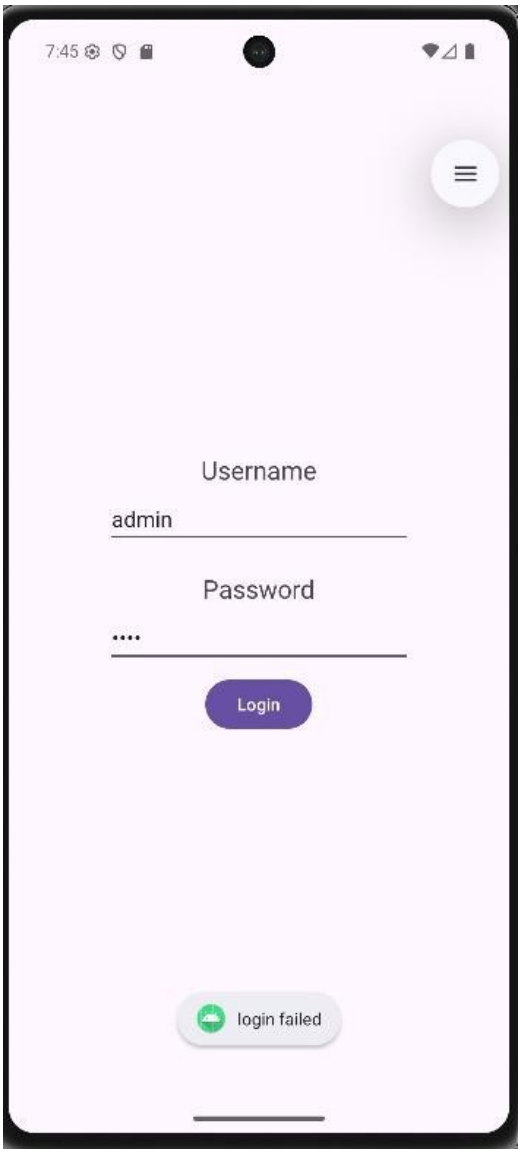
      btn.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            String username= user.getText().toString();
            String pas=password.getText().toString();

            if(username.equals("admin") && pas.equals("admin"))
               { Toast.makeText(MainActivity.this, "login successfully",
  Toast.LENGTH_SHORT).show();
            }else {
               Toast.makeText(MainActivity.this,"login
  failed",Toast.LENGTH_SHORT).show();
            }
```

```
            }
        });
    }
}
```

OUTPUT


7:45
Username
admin
Password
.....
Login
login successfully


7:45
Username
admin
Password
....
Login
login failed

EXPERIMENT NO 2

AIM:

Write a program that demonstrates Activity Lifecycle.

SOURCE CODE

Activiy_Main.Xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
  xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Activity Lifecycle"
      android:textSize="20sp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintHorizontal_bias="0.467"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.276" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.activity_2;
import android.os.Bundle;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        Log.i("state","onCreate");
    }
    protected void onStart(){
        super.onStart();
        Log.i("state","onStart");
    }
    protected void onResume(){
        super.onResume();
        Log.i("state","onResume");
    }
    protected void onPause(){
        super.onPause();
        Log.i("state","onPause");
    }
    protected void onStop(){
        super.onStop();
        Log.i("state","onStop");
    }
    protected void onRestart(){
        super.onRestart();
        Log.i("state","onRestart");
    }
    @Override
    protected void onDestroy(){
        super.onDestroy();
        Log.i("state","onRestart");
    }
}
```

OUTPUT

```
2025-10-25 07:58:30.035  7294-7294  DesktopModeFlags        com.example.activity_2  D  Toggle override initialized to: OVERRIDE_UNSET
2025-10-25 07:58:30.153  7294-7294  mple.activity_2         com.example.activity_2  I  hiddenapi: Accessing hidden method Landroid/view/ViewGroup;->makeOptionalFit
2025-10-25 07:58:30.309  7294-7294  state                   com.example.activity_2  I  onCreate
2025-10-25 07:58:30.443  7294-7294  state                   com.example.activity_2  I  onStart
2025-10-25 07:58:30.467  7294-7294  state                   com.example.activity_2  I  onResume
2025-10-25 07:58:30.553  7294-7294  HWUI                    com.example.activity_2  W  Unknown dataspace 0
2025-10-25 07:58:31.032  7294-7294  InsetsController        com.example.activity_2  D  hide(ime(), fromIme=false)
2025-10-25 07:58:31.032  7294-7294  ImeTracker              com.example.activity_2  I  com.example.activity_2:b0dcf66c: onCancelled at PHASE_CLIENT_ALREADY_HIDDEN
2025-10-25 07:58:32.951  7294-7303  mple.activity_2         com.example.activity_2  I  Compiler allocated 5042KB to compile void android.view.ViewRootImpl.performT
2025-10-25 07:58:34.459  7294-7337  ProfileInstaller        com.example.activity_2  D  Installing profile for com.example.activity_2
2025-10-25 08:00:04.767  7294-7294  state                   com.example.activity_2  I  onPause
2025-10-25 08:00:04.774  7294-7294  VRI[MainActivity]       com.example.activity_2  D  visibilityChanged oldVisibility=true newVisibility=false
2025-10-25 08:00:04.848  7294-7294  state                   com.example.activity_2  I  onStop
2025-10-25 08:00:12.522  7294-7294  ViewRootImpl            com.example.activity_2  D  Skipping stats log for color mode
2025-10-25 08:00:12.522  7294-7294  state                   com.example.activity_2  I  onRestart
2025-10-25 08:00:12.523  7294-7294  state                   com.example.activity_2  I  onStart
2025-10-25 08:00:12.525  7294-7294  state                   com.example.activity_2  I  onResume
2025-10-25 08:00:12.898  7294-7294  InsetsController        com.example.activity_2  D  hide(ime(), fromIme=false)
2025-10-25 08:00:12.898  7294-7294  ImeTracker              com.example.activity_2  I  com.example.activity_2:5cfca5f9: onCancelled at PHASE_CLIENT_ALREADY_HIDDEN
2025-10-25 08:00:42.206  7294-7294  state                   com.example.activity_2  I  onPause
2025-10-25 08:00:42.214  7294-7294  state                   com.example.activity_2  I  onStop
2025-10-25 08:00:42.595  7294-7294  WindowOnBackDispatcher  com.example.activity_2  W  sendCancelIfRunning: isInProgress=false callback=android.app.Activity$$Exter
2025-10-25 08:00:42.601  7294-7294  state                   com.example.activity_2  I  onDestroy
```

EXPERIMENT NO 3

AIM:

Implementing basic arithmetic operations of a simple calculator

SOURCE CODE

Activity_Main.Xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

    <TextView
       android:id="@+id/result"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Result"
       android:textSize="24sp"
       app:layout_constraintBottom_toBottomOf="parent"
       app:layout_constraintEnd_toEndOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintVertical_bias="0.153" />

    <EditText
       android:id="@+id/num1"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginStart="100dp"
       android:layout_marginTop="196dp"
       android:ems="10"
       android:inputType="text"
       android:hint="first number"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent" />
    <EditText
       android:id="@+id/num2"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginStart="100dp"
       android:layout_marginTop="268dp"
```

```
        Android:ems="10"
        android:inputType="text"
        android:hint="second"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/add"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="8dp"
        android:layout_marginTop="344dp"
        android:text="Add"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/sub"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="108dp"
        android:layout_marginTop="344dp"
        android:text="Sub"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/mul"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="208dp"
        android:layout_marginTop="344dp"
        android:text="mult"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/div"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="308dp"
        android:layout_marginTop="344dp"
        android:text="Div"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.calculator;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

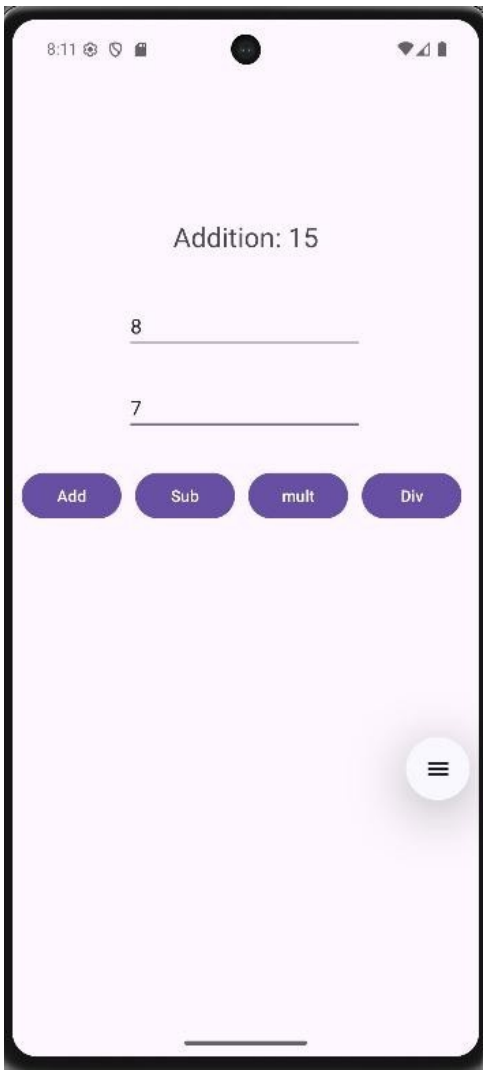
public class MainActivity extends AppCompatActivity
    { EditText first,second;
    Button btn1,btn2,btn3,btn4;
    TextView res;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        first=findViewById(R.id.num1);
        second=findViewById(R.id.num2);
        btn1=findViewById(R.id.add);
        btn2=findViewById(R.id.sub);
        btn3=findViewById(R.id.mul);
        btn4=findViewById(R.id.div);
        res=findViewById(R.id.result);

        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int n1=Integer.parseInt(first.getText().toString());
                int n2 = Integer.parseInt(second.getText().toString());
                int result = n1+n2;
                res.setText(String.valueOf("Addition: "+result));
            }
        });

        btn2.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int n1=Integer.parseInt(first.getText().toString());
                int n2 = Integer.parseInt(second.getText().toString());
                int result = n1-n2;
                res.setText(String.valueOf("sub:  "+result));
            }
```

```
});
      btn3.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            int n1=Integer.parseInt(first.getText().toString());
            int n2 = Integer.parseInt(second.getText().toString());
            int result = n1*n2;
            res.setText(String.valueOf("mult:  "+result));
         } });
      btn4.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            int n1=Integer.parseInt(first.getText().toString());
            int n2 = Integer.parseInt(second.getText().toString());
            int result = n1/n2;
            res.setText(String.valueOf("division: "+result));
         }
      });
   }}
```

OUTPUT



EXPERIMENT NO 4

AIM:

Implement validations on various UI controls.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true">
    <LinearLayout
        android:orientation="vertical"
        android:padding="16dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingStart="20sp"
        android:paddingEnd="20sp">
        <TextView
            android:text="Validaion form"
            android:textSize="20sp"
            android:textColor="@color/black"
            android:layout_gravity="center_horizontal"
            android:layout_marginBottom="10sp"
            android:layout_marginTop="12dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textStyle="bold"/>
        <EditText
            android:id="@+id/etName"
            android:hint="Name"
            android:inputType="textPersonName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content" />
        <EditText
            android:id="@+id/etEmail"
            android:hint="Email"
            android:inputType="textEmailAddress"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp" />
        <TextView
            android:text="Gender"
            android:textSize="16sp"
            android:layout_marginTop="12dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
```

```
<RadioGroup
        android:id="@+id/rgGender"
        android:orientation="horizontal"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">
        <RadioButton
            android:id="@+id/rbMale"
            android:text="Male"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"  />
        <RadioButton
            android:id="@+id/rbFemale"
            android:text="Female"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="20dp"  />
    </RadioGroup>
    <EditText
        android:id="@+id/etMobile"
        android:hint="Mobile Number"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp" />
    <EditText
        android:id="@+id/etDob"
        android:hint="Date of Birth"
        android:focusable="false"
        android:clickable="true"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp" />
    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp" />
    <EditText
        android:id="@+id/etConfirmPassword"
        android:hint="Confirm Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp" />
    <TextView
        android:text="State"
```

```
        android:textSize="16sp"
        android:layout_marginTop="12dp"
        android:layout_width="wrap_content"
          android:layout_height="wrap_content"/>
    <Spinner
        android:id="@+id/spinnerState"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"  />
    <TextView
        android:text="Hobbies"
        android:textSize="16sp"
        android:layout_marginTop="12dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <CheckBox
        android:id="@+id/cbReading"
        android:text="Reading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"  />
    <CheckBox
        android:id="@+id/cbTraveling"
        android:text="Traveling"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />
    <CheckBox
        android:id="@+id/cbGaming"
        android:text="Gaming"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"  />
    <Button
        android:id="@+id/btnSubmit"
        android:text="Submit"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp" />
  </LinearLayout>
</ScrollView>
```

ActivityMain.java

```
package com.example.validation;
import android.app.DatePickerDialog;
import android.os.Bundle;
import android.text.TextUtils;
import android.util.Patterns;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
```

```
import java.util.Calendar;
public class MainActivity extends AppCompatActivity {
   EditText Name, Email, Mobile, Dob, Password, ConfirmPassword;

  RadioGroup Gender;
   Spinner spinnerState;
   CheckBox Reading, Traveling, Gaming;
   Button btnSubmit;
   String[] states = {"Select State", "Maharashtra", "Gujarat", "Karnataka","kerala"};
   String[] district={"select district","malappuram","palakkad","alappuzha","kozhikode"};

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      Name = findViewById(R.id.etName);
      Email = findViewById(R.id.etEmail);
      Mobile = findViewById(R.id.etMobile);
      Dob = findViewById(R.id.etDob);
      Password = findViewById(R.id.etPassword);
      ConfirmPassword = findViewById(R.id.etConfirmPassword);
      Gender = findViewById(R.id.rgGender);
      spinnerState = findViewById(R.id.spinnerState);
      Reading = findViewById(R.id.cbReading);
      Traveling = findViewById(R.id.cbTraveling);
      Gaming = findViewById(R.id.cbGaming);
      btnSubmit = findViewById(R.id.btnSubmit);

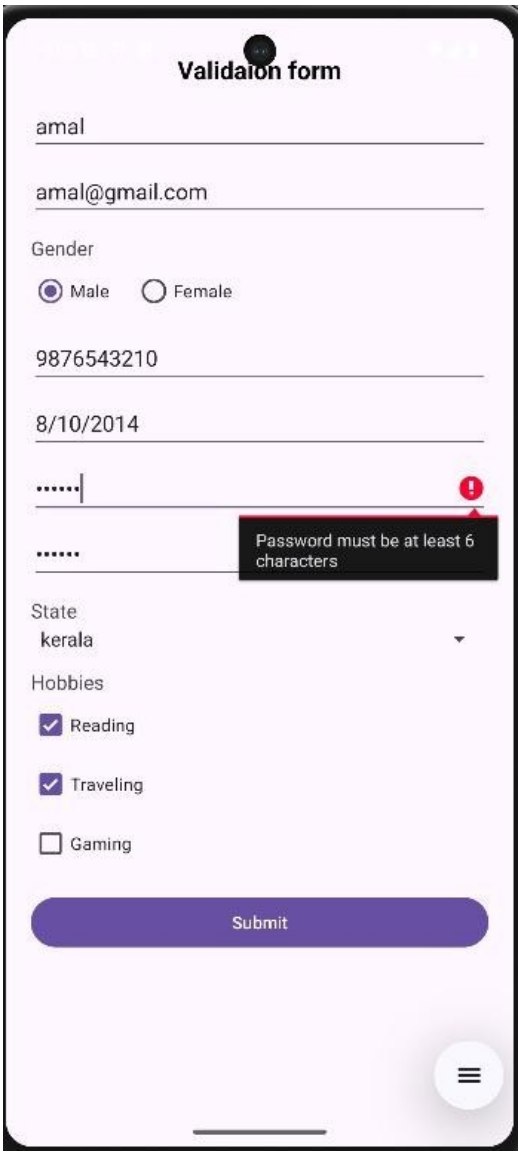
      ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
  android.R.layout.simple_spinner_item, states);
      adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
      spinnerState.setAdapter(adapter);

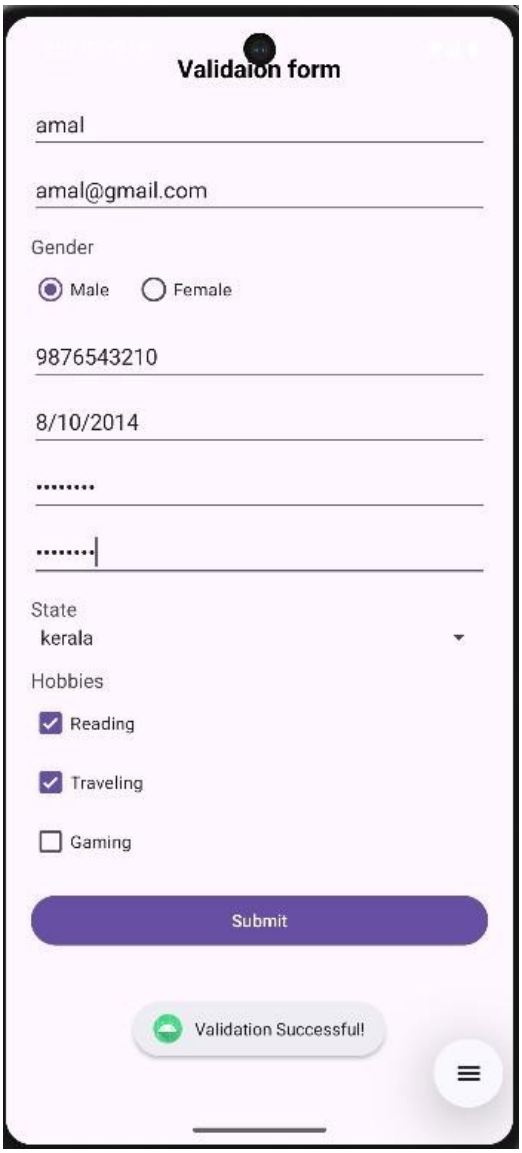
      Dob.setOnClickListener(view -> {
         Calendar calendar = Calendar.getInstance();
         int y = calendar.get(Calendar.YEAR);
         int m = calendar.get(Calendar.MONTH);
         int d = calendar.get(Calendar.DAY_OF_MONTH);
         new DatePickerDialog(MainActivity.this, (view1, year, month, dayOfMonth) ->
            { Dob.setText(dayOfMonth + "/" + (month + 1) + "/" + year);
         }, y, m, d).show();
      });
      btnSubmit.setOnClickListener(view -> validateForm());
   }
   private void validateForm() {
      String name = Name.getText().toString().trim();
      String email = Email.getText().toString().trim();
      String mobile = Mobile.getText().toString().trim();
      String dob = Dob.getText().toString().trim();
```

```
        String password = Password.getText().toString();
        String confirmPassword = ConfirmPassword.getText().toString();

        if (name.isEmpty() || !name.matches("[a-zA-Z ]+"))
            { Name.setError("Enter a valid name");
            return;
    }
        if (!Patterns.EMAIL_ADDRESS.matcher(email).matches())
            { Email.setError("Enter a valid email");
            return;
        }
        if (Gender.getCheckedRadioButtonId() == -1) {
            Toast.makeText(this, "Select gender", Toast.LENGTH_SHORT).show();
            return;
        }
        if (mobile.length() != 10 || !TextUtils.isDigitsOnly(mobile))
            { Mobile.setError("Enter valid 10-digit mobile number");
            return;
        }
        if (dob.isEmpty())
            { Dob.setError("Select date of
            birth"); return;
        }
        if (password.length() < 8) {
            Password.setError("Password must be at least 6 characters");
            return;
        }
        if (confirmPassword.isEmpty() || !password.equals(confirmPassword))
            { ConfirmPassword.setError("Passwords do not match");
            return;
        }
        if (spinnerState.getSelectedItemPosition() == 0) {
            Toast.makeText(this, "Select a state", Toast.LENGTH_SHORT).show();
            return;
        }
        if (!Reading.isChecked() && !Traveling.isChecked() && !Gaming.isChecked())
            { Toast.makeText(this, "Select at least one hobby", Toast.LENGTH_SHORT).show();
            return;
        }
        Toast.makeText(this, "Validation Successful!", Toast.LENGTH_LONG).show();
    }
}
```

OUTPUT


Validaion form
amal
amal@gmail.com
Gender
Male
Female
9876543210
8/10/2014
Password must be at least 6 characters
State
kerala
Hobbies
Reading
Traveling
Gaming
Submit


Validaion form
amal
amal@gmail.com
Gender
Male
Female
9876543210
8/10/2014
State
kerala
Hobbies
Reading
Traveling
Gaming
Submit
Validation Successful!

EXPERIMENT NO 5

AIM:

Design a simple Calculator using GridLayout and Cascaded LinearLayout.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
       android:id="@+id/display"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:background="#EEE"
       android:gravity="end|center_vertical"
       android:padding="16dp"
       android:text="0"
       android:textSize="36sp"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintEnd_toEndOf="parent" />

    <EditText
       android:id="@+id/num1"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:ems="10"
       android:hint="enter the first number"
       android:inputType="text"
       android:background="#E0E0E0"
       android:padding="16dp"
       android:layout_marginTop="10dp"
       app:layout_constraintTop_toBottomOf="@id/display"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintEnd_toEndOf="parent" />
    <EditText
       android:id="@+id/num2"
       android:layout_width="0dp"
       android:layout_height="wrap_content"
```

```
        android:ems="10"
    android:hint="enter the second
    number"
android:inputType="text"
    android:background="#E0E0E0"
    android:padding="16dp"
    android:layout_marginTop="8dp"
    app:layout_constraintTop_toBottomOf="@id/num1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />
<GridLayout
    android:id="@+id/buttonGrid"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:alignmentMode="alignMargins"
    android:columnCount="4"
    android:rowCount="2"
    android:useDefaultMargins="true"
    android:layout_marginTop="16dp"
    app:layout_constraintTop_toBottomOf="@id/num2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent">
    <Button
        android:id="@+id/buttonAdd"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="+" />
    <Button
        android:id="@+id/buttonSubtract"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="-" />
    <Button
        android:id="@+id/buttonMultiply"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="×" />
    <Button
        android:id="@+id/buttonDivide"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_columnWeight="1"
        android:text="÷" />
    <Button
        android:id="@+id/buttonReset"
        android:layout_width="0dp"
```

```
            android:layout_height="wrap_content"
            android:layout_columnSpan="2"
            android:layout_columnWeight="1"
            android:text="C" />
           <Button
            android:id="@+id/buttonEquals"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_columnSpan="2"
            android:layout_columnWeight="1"
            android:text="=" />
    </GridLayout>
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

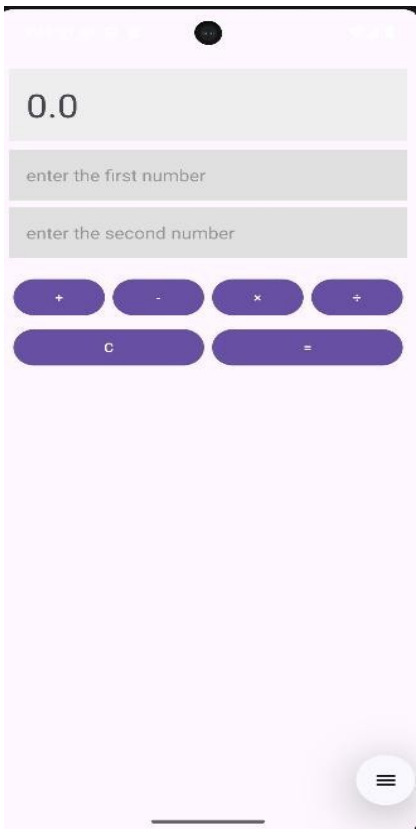
```
package com.example.calculator_grid;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

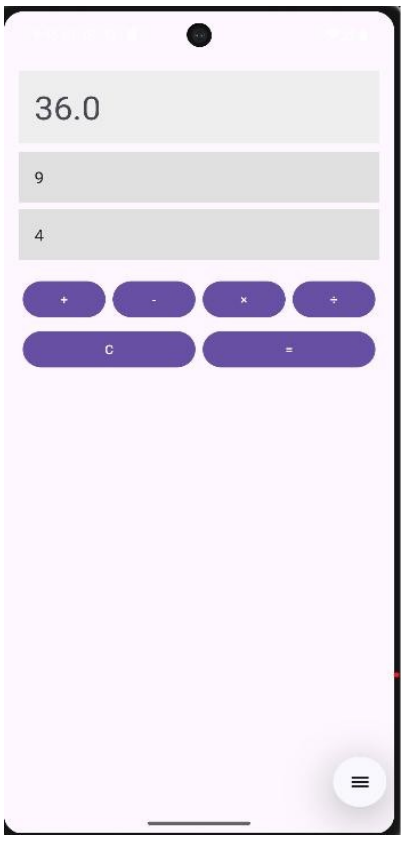
public class MainActivity extends AppCompatActivity
    { EditText num1, num2;
    TextView display;
    Button add, subtract, multiply, divide, equals, reset;
    String operator = "";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        num1 = findViewById(R.id.num1);
        num2 = findViewById(R.id.num2);
        display = findViewById(R.id.display);
        add = findViewById(R.id.buttonAdd);
        subtract = findViewById(R.id.buttonSubtract);
        multiply = findViewById(R.id.buttonMultiply);
        divide = findViewById(R.id.buttonDivide);
        equals = findViewById(R.id.buttonEquals);
        reset = findViewById(R.id.buttonReset);
        add.setOnClickListener(v -> operator = "+");
        subtract.setOnClickListener(v -> operator =
        "-");
        multiply.setOnClickListener(v -> operator = "*");
        divide.setOnClickListener(v -> operator = "/");

        equals.setOnClickListener(v -> {
            double n1 = Double.parseDouble(num1.getText().toString());
```

```
        double n2 = Double.parseDouble(num2.getText().toString());
        double result = 0;
        switch (operator)
            { case "+":
                result = n1 + n2;
                break;
            case "-":
                result = n1 - n2;
                break;
            case "*":
                result = n1 * n2;
                break;
            case "/":
                result = n1 / n2;
                break;
            default:
                display.setText("Select operator");
                return;
        }
        display.setText(String.valueOf(result));  });
    reset.setOnClickListener(v -> {
        num1.setText("");
        num2.setText("");
        display.setText("0.0");
        operator = "";
    }); } }
```

OUTPUT





EXPERIMENT NO 6

AIM:

Create a Facebook page using RelativeLayout; set properties using .xml file

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:padding="20sp"
   android:background="#1877F2">

   <TextView
      android:id="@+id/logo"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="facebook"
      android:textSize="36sp"
      android:textStyle="bold"
      android:textColor="@color/white"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="60sp"/>
   <EditText
      android:id="@+id/email"
      android:layout_width="wrap_content"
      android:layout_height="50dp"
      android:ems="10"
      android:inputType="textEmailAddress"
      android:hint="Email id"
      android:background="@color/white"
      android:padding="10sp"
      android:layout_below="@id/logo"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="180dp"/>
   <EditText
      android:id="@+id/pass"
      android:layout_width="wrap_content"
      android:layout_height="50dp"
      android:ems="10"
      android:inputType="textPassword"
      android:padding="10sp"
      android:hint="password"
      android:layout_marginTop="40dp"
      android:background="@color/white"
```

```
      android:layout_centerHorizontal="true"
       android:layout_below="@id/email"/>
   <Button
       android:id="@+id/button"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Button"
       android:layout_below="@id/pass"
       android:scaleX="1.2"
       android:scaleY="1.2"
       android:layout_centerHorizontal="true"
       android:layout_marginTop="30sp"/>
</RelativeLayout>
```

MainActivity.java

```
package com.example.facebook;

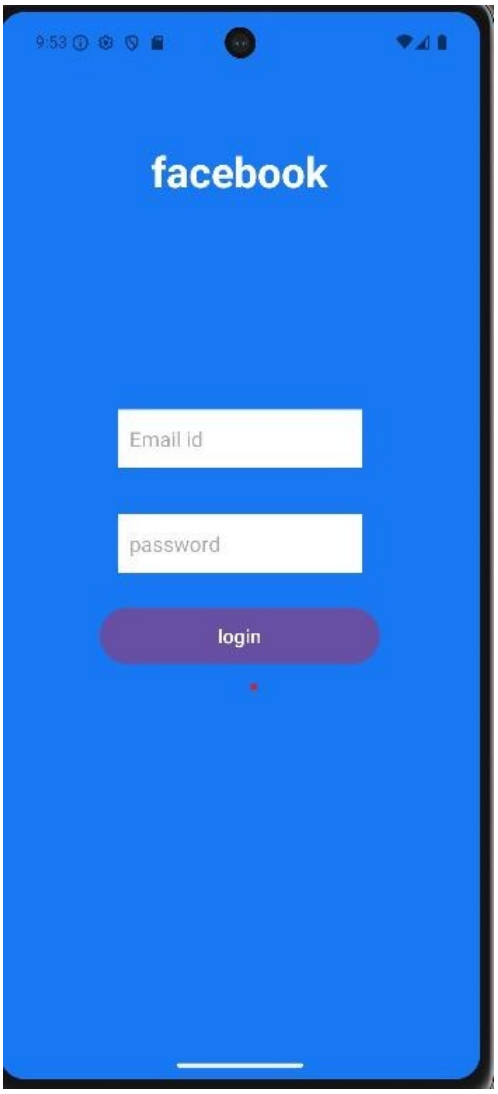
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity
   { EditText user,password;
   Button btn;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       EdgeToEdge.enable(this);

       user=findViewById(R.id.email);
       password=findViewById(R.id.pass);
       btn=findViewById(R.id.button);
       btn.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               String mail= user.getText().toString();
               String pass=password.getText().toString();
               if(mail.equals("user@gmail.com") && pass.equals("user123"))
{ Toast.makeText(MainActivity.this, "login successfully", Toast.LENGTH_SHORT).show();
       }
```

```
else {
Toast.makeText(MainActivity.this, "login failed", Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

OUTPUT





EXPERIMENT NO 7

AIM:
Implement Adapters and perform Exception Handling

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20sp">

    <TextView
        android:id="@+id/head"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="70dp"
        android:text="Countries"
        android:textSize="30dp"
         />
    <ListView
        android:id="@+id/listview"
        android:layout_width="wrap_content"
        android:layout_below="@id/head"
        android:layout_height="wrap_content"
        android:layout_marginTop="20sp"
        android:divider="@color/cardview_dark_background"
        android:dividerHeight="1dp"
         />
</RelativeLayout>
```

MainActivity.java

```
package com.example.adapter;

import android.os.Bundle;
import android.util.AndroidException;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity
    { ListView list;
    String[] countries={"usa","india","canada","uae","portugal"};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        list=findViewById(R.id.listview);
        try {
            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                    this,
                    android.R.layout.simple_list_item_1,
                    countries  );
            list.setAdapter(adapter);
        } catch (Exception e){
            Toast.makeText(this, e.getMessage(), Toast.LENGTH_SHORT).show();
        }
    }
}
```

OUTPUT



EXPERIMENT NO 8

AIM:
Implement Intent to navigate between multiple activities.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

    <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Implicit Intent"
       android:textSize="30dp"
       app:layout_constraintBottom_toBottomOf="parent"
       app:layout_constraintEnd_toEndOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintVertical_bias="0.127" />

    <Button
       android:id="@+id/button"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginStart="156dp"
       android:layout_marginTop="256dp"
       android:text="Google"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

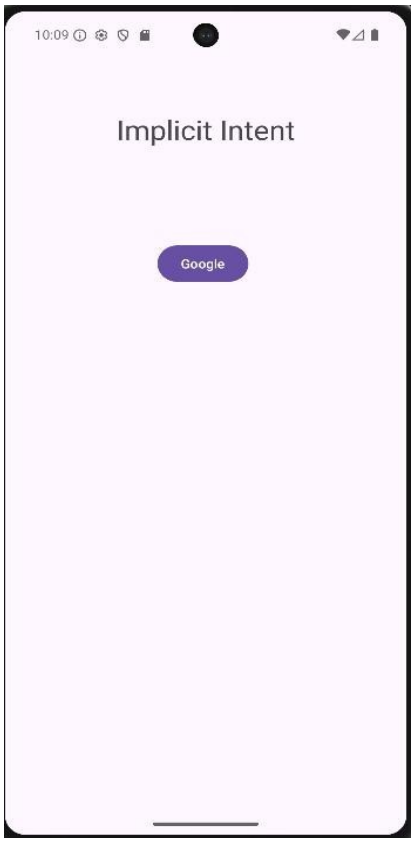
MainActivity.java

```
package com.example.implicitintent;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
```

```
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity
    { Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        btn=findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent i = new Intent(Intent.ACTION_VIEW);
                i.setData(Uri.parse("https://www.google.com/"));
                startActivity(i);
            }
        });
    }
}
```

OUTPUT





EXPERIMENT NO 9

AIM:
Develop an application that works with Explicit Intents.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
    <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Explicit intent"
       android:textSize="25dp"
       app:layout_constraintBottom_toBottomOf="parent"
       app:layout_constraintEnd_toEndOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintVertical_bias="0.141" />
    <Button
       android:id="@+id/btn"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginStart="148dp"
       android:layout_marginTop="272dp"
       android:text="Click Here"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.intent;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

```
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity
   { Button button;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      button=findViewById(R.id.btn);
      button.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            Intent i = new Intent(MainActivity.this,MainActivity2.class);
            startActivity(i);
         }
      });
   } }
```

Activity_Main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity2">

    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Second Activity"
      android:textSize="25dp"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.141" />
   <Button
      android:id="@+id/btn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:layout_marginStart="148dp"
      android:layout_marginTop="272dp"
```

```
        android:text="Back"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

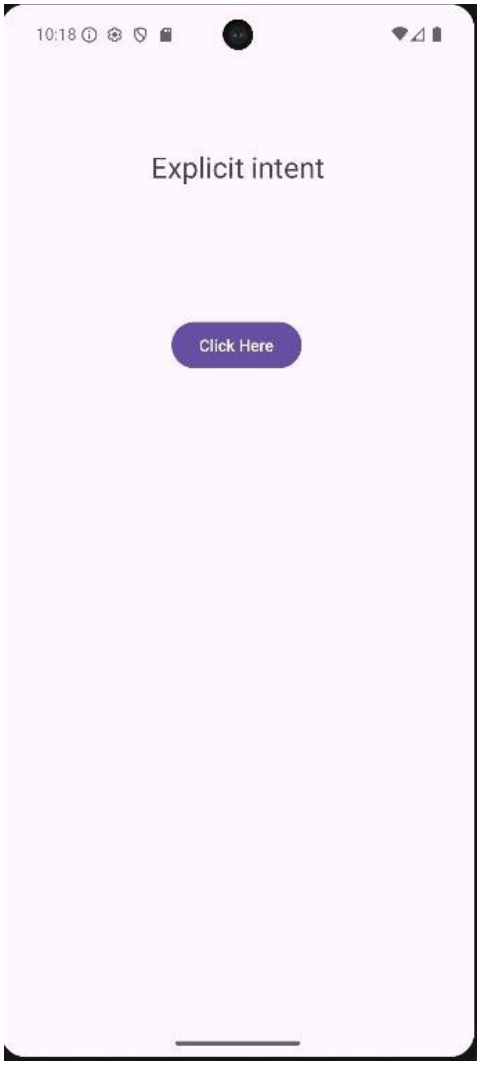
MainActivity2.java

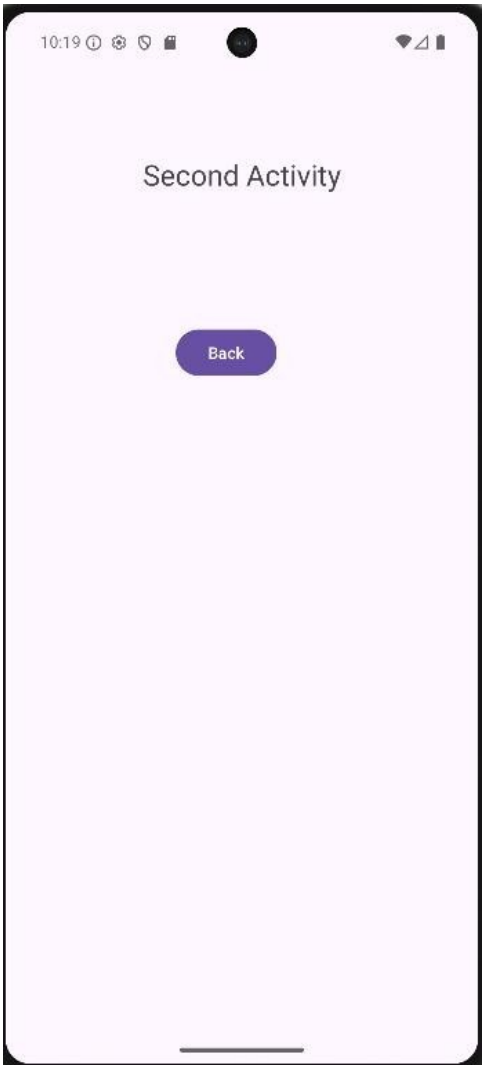
```
package com.example.intent;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity2 extends AppCompatActivity
    { Button btn1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main2);
        btn1=findViewById(R.id.btn);
        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent i = new Intent(MainActivity2.this,MainActivity.class);
                startActivity(i);
            }
        });
    }
}
```

OUTPUT





EXPERIMENT NO 10

AIM:
Develop an application that implements Spinner component.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

    <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Select a Country"
       android:textSize="20sp"
       app:layout_constraintBottom_toBottomOf="parent"
       app:layout_constraintEnd_toEndOf="parent"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintVertical_bias="0.162" />
    <Spinner
       android:id="@+id/countrySpinner"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginTop="188dp"
       app:layout_constraintEnd_toEndOf="parent"
       app:layout_constraintHorizontal_bias="0.498"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.spinner;

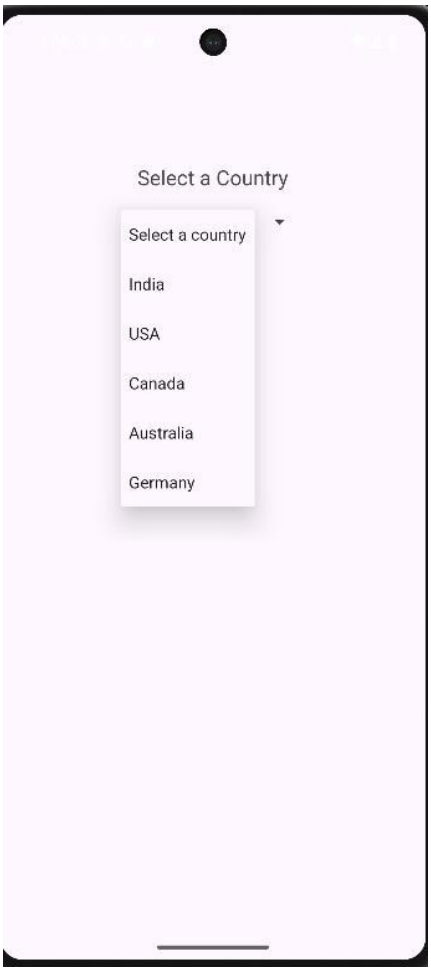
import  android.os.Bundle;
import android.view.View;
import  android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
```

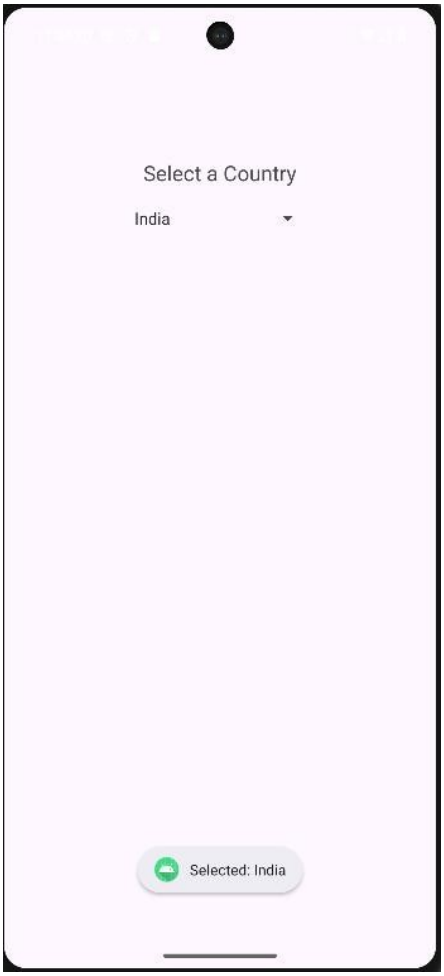
```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    Spinner countrySpinner;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        countrySpinner = findViewById(R.id.countrySpinner);
        String[] countries = {"Select a country", "India", "USA", "Canada", "Australia",
   "Germany"};
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
         this,
        android.R.layout.simple_spinner_item, countries);
        adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
        countrySpinner.setAdapter(adapter);

    countrySpinner.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {
      @Override
      public void onItemSelected(AdapterView<?> parent, View view, int position, long id)
            { if (position != 0) {
           String selectedCountry = parent.getItemAtPosition(position).toString();
             Toast.makeText(MainActivity.this, "Selected: " + selectedCountry,
                  Toast.LENGTH_SHORT).show();
               }
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent)
                    { Toast.makeText(MainActivity.this, "No selection made",
                Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

OUTPUT


Select a Country
Select a country
India
USA
Canada
Australia
Germany


Select a Country
India
Selected: India

-**EXPERIMENT NO 11**

AIM:
Develop an application that toggles image using FrameLayout.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity"
    android:padding="27dp">

    <FrameLayout
       android:id="@+id/frameLayout"
       android:layout_width="250dp"
       android:layout_height="250dp"
       android:layout_marginTop="26dp"
       app:layout_constraintTop_toTopOf="parent"
       app:layout_constraintBottom_toTopOf="@id/toggleButton"
       app:layout_constraintStart_toStartOf="parent"
       app:layout_constraintEnd_toEndOf="parent">
       <ImageView
          android:id="@+id/image1"
          android:layout_width="match_parent"
          android:layout_height="match_parent"
          android:src="@drawable/images"
          android:scaleType="centerCrop"
          android:visibility="visible" />
       <ImageView
          android:id="@+id/image2"
          android:layout_width="match_parent"
          android:layout_height="match_parent"
          android:src="@drawable/car1"
          android:scaleType="centerCrop"
          android:visibility="gone" />
   </FrameLayout>

   <Button
       android:id="@+id/toggleButton"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Toggle Image"
```

```
        app:layout_constraintTop_toBottomOf="@id/frameLayout"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.frame_layout;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
   { ImageView image1, image2;
   Button toggleButton;
   boolean showingFirst = true;

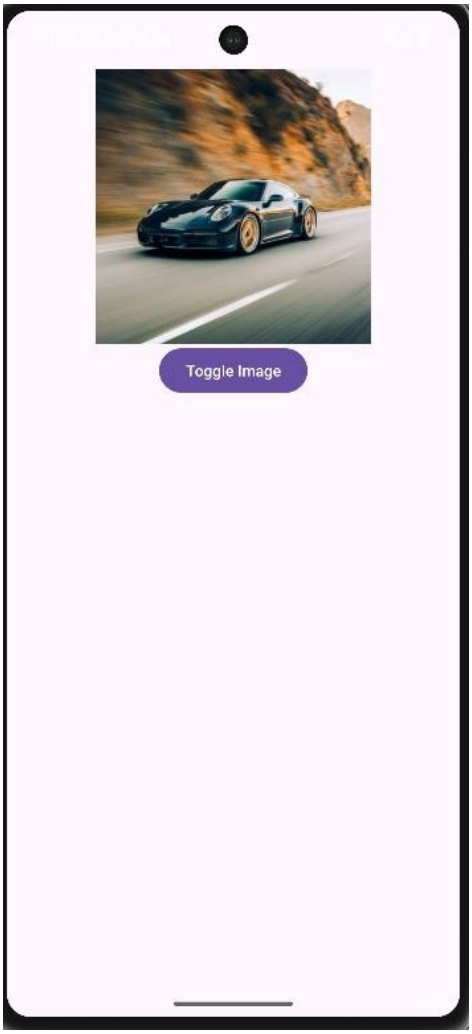
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

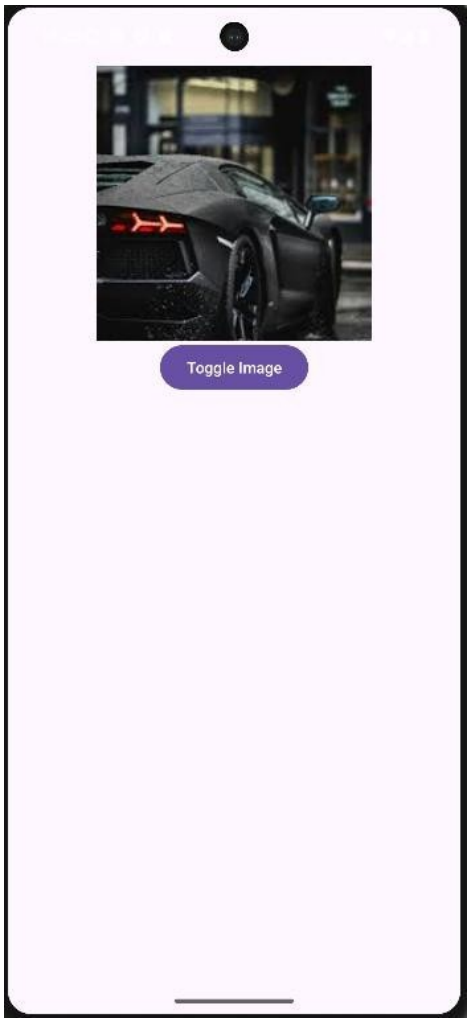
      image1 = findViewById(R.id.image1);
      image2 = findViewById(R.id.image2);
      toggleButton = findViewById(R.id.toggleButton);

      toggleButton.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View v)
            { if (showingFirst) {
               image1.setVisibility(View.GONE);
               image2.setVisibility(View.VISIBLE);
            } else {
               image1.setVisibility(View.VISIBLE);
               image2.setVisibility(View.GONE);

            }
         }  showingFirst = !showingFirst;
      });
   }
}
```

OUTPUT





EXPERIMENT NO 12

AIM:
Develop an application that uses ArrayAdapter with ListView.

SOURCE CODE

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/headingText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:paddingBottom="10dp"
        android:text="Country List"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginTop="50dp"
        android:layout_centerHorizontal="true"
        android:paddingTop="10sp"/>
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:divider="@color/cardview_dark_background"
        android:dividerHeight="1dp"
        android:layout_below="@+id/headingText"
        android:layout_marginTop="10dp"
        android:padding="10sp"/>
</RelativeLayout>
```

MainActivity.java

```
package com.example.adapter_10noexception;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```
 ListView listView;
    String[] countries = {"USA", "India", "China", "Germany", "France", "Australia",
   "Canada"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);
            ArrayAdapter<String> adapter = new ArrayAdapter<>(
                    this,
                    android.R.layout.simple_list_item_1, countries);
            listView.setAdapter(adapter);
    }
}
```

OUTPUT

